

Aerodynamics, Turbulence, and Computational Fluid Dynamics

A physics-first introduction, with full step-by-step derivations

How to Use This Handout

Every boxed result below is derived, not just stated. Alongside each mathematical step is a short note explaining *what* operation is being performed and *why* it's the right move at that point – the goal is that you could reproduce the derivation yourself after reading it once, not just recognize the final formula.

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1 The Fluid as a Continuum

Air at standard conditions contains roughly 10^{25} molecules per cubic meter. Tracking each one is neither feasible nor useful for engineering purposes, so instead of a discrete molecular description we replace it with continuous averaged fields:

$$\rho(\mathbf{x}, t), \quad p(\mathbf{x}, t), \quad \mathbf{u}(\mathbf{x}, t), \quad T(\mathbf{x}, t).$$

Each of these is a local average taken over a volume that is large compared to the spacing between molecules, but small compared to the smallest flow feature we care about (a boundary-layer thickness, a turbulent eddy). The formal condition for this averaging to make sense is that the Knudsen number

$$Kn = \frac{\lambda}{L}$$

be small, where λ is the mean free path between molecular collisions and L is that smallest flow length scale. When $Kn \ll 1$, the continuum approximation holds and we can write down PDEs for smooth fields instead of tracking particles – this is the foundation everything else in this handout is built on.

1.1 Eulerian vs. Lagrangian Descriptions

There are two equally valid ways to describe the same flow. **Eulerian:** park yourself at a fixed point in space and record how the flow properties there change in time, $\mathbf{u} = \mathbf{u}(\mathbf{x}, t)$. **Lagrangian:** tag an individual fluid parcel and follow it around, recording $\mathbf{x}(t; \mathbf{x}_0)$, the trajectory of the parcel that started at $\mathbf{x}_0$, governed by

$$\frac{d\mathbf{x}}{dt} = \mathbf{u}(\mathbf{x}(t), t), \quad \mathbf{x}(0) = \mathbf{x}_0.$$

CFD is built on the Eulerian view because it lets us store field values on a fixed mesh rather than track the identity of every deforming parcel. But Newton's second law is naturally a Lagrangian statement (it applies to a piece of matter, not a point in space), so we'll need a way to translate between the two – that's exactly what the material derivative in Section 3.2 does.

2 Conservation of Mass

2.1 Setting up the balance

The physical statement is simple: mass is neither created nor destroyed. To turn that into an equation, draw an imaginary, fixed control volume V in space, with boundary surface ∂V and outward unit normal $\mathbf{n}$. The total mass inside is $\int_V \rho dV$. The only way that quantity can change is if mass physically crosses the boundary ∂V , so

$$\frac{d}{dt} \int_V \rho dV = - \oint_{\partial V} \rho \mathbf{u} \cdot \mathbf{n} dA.$$

The right-hand side is (minus) the flux of mass leaving through the surface: $\rho \mathbf{u} \cdot \mathbf{n}$ is the mass flux normal to a patch of surface, and integrating it over the whole closed surface gives the net outflow.

2.2 Converting the surface integral to a volume integral

We want a statement that holds at every *point*, not just integrated over some arbitrary volume, so we need to get rid of the surface integral. The divergence theorem lets us do exactly that:

$$\oint_{\partial V} \rho \mathbf{u} \cdot \mathbf{n} dA = \int_V \nabla \cdot (\rho \mathbf{u}) dV.$$

Substituting this in and moving everything to one side,

$$\int_V \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) \right] dV = 0.$$

2.3 From integral to differential form

This integral is zero for *every* choice of control volume V , including arbitrarily small ones shrunk down around any point. The only way an integral of a continuous function can vanish over every possible sub-region is if the function itself is zero everywhere. That localization argument is what lets us drop the integral sign and write the pointwise PDE:

$$\boxed{\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0.} \quad (2.1)$$

2.4 Specializing to incompressible flow

Expand the divergence term with the product rule, $\nabla \cdot (\rho \mathbf{u}) = \mathbf{u} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{u}$, and regroup the two terms involving ρ using the material derivative introduced formally in Section 3.2:

$$\frac{\partial \rho}{\partial t} + \mathbf{u} \cdot \nabla \rho + \rho \nabla \cdot \mathbf{u} = 0 \quad \Longleftrightarrow \quad \frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0. \quad (2.2)$$

$D\rho/Dt$ is the rate of change of density *following a fluid parcel*. If the fluid is incompressible, a parcel's density never changes as it moves, so $D\rho/Dt = 0$ by definition of "incompressible." Setting that term to zero in (2.2) leaves

$$\boxed{\nabla \cdot \mathbf{u} = 0,} \quad (2.3)$$

or in components, $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$. Geometrically, the divergence of the velocity field measures the fractional rate at which a small parcel's volume is expanding, $\dot{V}/V = \nabla \cdot \mathbf{u}$; equation (2.3) is just the statement that this rate is zero, i.e. parcels can deform and stretch but never change volume.

3 Conservation of Momentum

3.1 Introducing the stress tensor

Newton's second law says force equals mass times acceleration. To apply it to a fluid parcel we first need to know what forces act on its surface. Cauchy's stress theorem states that no matter how a surface element inside the fluid is oriented, the force per unit area (traction) on it depends linearly on the orientation:

$$t_i = \sigma_{ij}n_j,$$

where σ_{ij} is the stress tensor and n_j is the surface normal. Applying $\mathbf{F} = m\mathbf{a}$ to an infinitesimal fluid element, and converting the surface force (an integral over the element's boundary) into a volume force via the divergence theorem exactly as we did for mass in Section 2.2, gives Cauchy's equation of motion, per unit volume:

$$\rho \frac{Du_i}{Dt} = \frac{\partial \sigma_{ij}}{\partial x_j} + \rho g_i. \quad (3.1)$$

This equation is completely general – it holds for any continuous material, solid or fluid – because we haven't yet said anything about *how* σ_{ij} relates to the motion of the fluid. That relation is the missing physical ingredient.

3.2 The constitutive law for a Newtonian fluid

We now supply that ingredient. A fluid at rest can only push, never shear, so the stress must reduce to an isotropic pressure, $\sigma_{ij} = -p\delta_{ij}$, when $\mathbf{u} = 0$. A Newtonian fluid is defined by assuming the *additional* stress produced by motion is linear in the local velocity gradients (the simplest possible assumption consistent with the fluid having no preferred direction):

$$\sigma_{ij} = -p\delta_{ij} + \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) + \lambda \delta_{ij} \frac{\partial u_k}{\partial x_k}. \quad (3.2)$$

μ is the dynamic viscosity; λ is the bulk viscosity, associated with volume changes. For incompressible flow we already showed $\partial u_k/\partial x_k = 0$ in (2.3), so the λ term drops out automatically.

3.3 Assembling the Navier–Stokes equation

Substitute the constitutive law (3.2) into Cauchy's equation (3.1). We need $\partial \sigma_{ij}/\partial x_j$: the pressure part contributes $-\partial p/\partial x_i$, and the viscous part contributes $\mu \partial^2 u_i/\partial x_j \partial x_j = \mu \nabla^2 u_i$ (the second term in the parentheses of (3.2) vanishes upon differentiating and using (2.3) again). Collecting terms:

$$\boxed{\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \rho \mathbf{g}.} \quad (3.3)$$

Nothing new has been assumed physically beyond Newton's second law and the linear (Newtonian) stress-strain relationship; this is simply what those two assumptions look like once written out in full.

3.4 The Material Derivative

We used Du_i/Dt above without deriving it – here's where it comes from. For any scalar or vector field f evaluated along a moving parcel's trajectory $\mathbf{x}(t)$, the chain rule gives

$$\frac{Df}{Dt} = \frac{d}{dt}f(\mathbf{x}(t), t) = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} = \frac{\partial f}{\partial t} + u_i \frac{\partial f}{\partial x_i},$$

using $dx_i/dt = u_i$, the definition of velocity. Applying this to each component of velocity itself gives the acceleration of a fluid parcel:

$$\boxed{\frac{D\mathbf{u}}{Dt} = \underbrace{\frac{\partial\mathbf{u}}{\partial t}}_{\text{local}} + \underbrace{\mathbf{u} \cdot \nabla \mathbf{u}}_{\text{convective}}} \quad (3.4)$$

The first term is how velocity at a fixed point changes with time; the second is the extra acceleration a parcel feels purely from moving to a new location where the (possibly steady) field has a different value. Concrete check: steady 1-D nozzle flow, $u = u(x)$ only, so $\partial u/\partial t = 0$ identically, yet a parcel clearly speeds up as it moves into the constriction. For $A(x) = A_0 e^{-x/L}$ with mass flux $Au = \text{const}$, continuity gives $u(x) = u_0 e^{x/L}$, so

$$\frac{Du}{Dt} = u \frac{du}{dx} = \frac{u_0^2}{L} e^{2x/L} \neq 0,$$

confirming the acceleration comes entirely from the convective term.

3.5 Reading off the pressure and viscous force terms

Going back to (3.2)–(3.3): the pressure force per unit volume is $-\nabla p$, which points from high pressure toward low pressure (the fluid is pushed away from regions where it's being squeezed harder), exactly analogous to $-\nabla V$ giving force from a potential energy landscape in particle mechanics. The viscous force per unit volume is $\mu \nabla^2 \mathbf{u}$; the Laplacian measures how different a point's value is from the average of its neighbours, so this term drives diffusive smoothing of the velocity field – physically, adjacent layers of fluid moving at different speeds drag on one another until the differences smooth out, exactly as heat diffuses from hot regions to cold ones. In simple shear this reduces to the familiar

$$\boxed{\tau = \mu \frac{\partial u}{\partial y}} \quad (3.5)$$

4 Nondimensionalization and Reynolds Number

4.1 Why nondimensionalize

The physical variables in (3.3) carry units, which makes it hard to compare the relative size of different terms directly (you can't compare "how big is a pressure gradient" to "how big is a viscous term" without a common yardstick). Nondimensionalizing removes the units and reveals which dimensionless *groups* of parameters actually control the flow's behaviour.

4.2 The rescaling

Pick a characteristic length L and velocity U for the problem (e.g. chord length and freestream speed), and define dimensionless variables using them as yardsticks:

$$\mathbf{x}^* = \frac{\mathbf{x}}{L}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U}, \quad t^* = \frac{tU}{L}, \quad p^* = \frac{p}{\rho U^2}.$$

Each starred quantity is now order-1 by construction. Substitute these definitions into (3.3) – for instance $\partial/\partial x = (1/L)\partial/\partial x^*$ and $\mathbf{u} = U\mathbf{u}^*$, so each term picks up a specific combination of U and L as a prefactor. Carrying this through term by term and then dividing the whole equation by the common prefactor $\rho U^2/L$ (the size of the convective term) gives

$$\boxed{\frac{\partial \mathbf{u}^*}{\partial t^*} + \mathbf{u}^* \cdot \nabla^* \mathbf{u}^* = -\nabla^* p^* + \frac{1}{Re} \nabla^{*2} \mathbf{u}^* + \frac{1}{Fr^2} \hat{\mathbf{g}}}, \quad (4.1)$$

with the dimensionless groups that popped out of the algebra defined as

$$\boxed{Re = \frac{\rho UL}{\mu} = \frac{UL}{\nu}}, \quad Fr = \frac{U}{\sqrt{gL}}.$$

4.3 Why this matters: dynamic similarity

Because (4.1) is written entirely in terms of $\mathbf{x}^*, t^*, \mathbf{u}^*, p^*$ and the single number Re (ignoring gravity for aerodynamic flows), two flows around geometrically similar shapes that share the same value of Re satisfy the *identical* governing equation with identical boundary conditions – so their dimensionless solutions $\mathbf{u}^*, p^*$ must be identical too. This is exactly why testing a small-scale wind-tunnel model at matched Re tells you about the full-size aircraft. Re itself is just the ratio of the two competing terms on the right of (4.1): it tells you which physical effect (inertia or viscosity) dominates the force balance, not directly how turbulent the flow is.

5 Pressure, Lift, and Drag

5.1 Building the force from surface tractions

Going back to the traction $t_i = \sigma_{ij}n_j$ from Section 3.1, and splitting it into its pressure and viscous pieces as in (3.2), the total aerodynamic force and moment are just the surface integrals of that traction over the body:

$$\mathbf{F} = \oint_S (-p\mathbf{n} + \boldsymbol{\tau}) dS, \quad \mathbf{M} = \oint_S \mathbf{r} \times (-p\mathbf{n} + \boldsymbol{\tau}) dS,$$

with $\mathbf{r}$ measured from whatever moment reference point you choose (commonly the quarter-chord). This is nothing more than summing up the pressure push and viscous drag over every patch of the surface.

5.2 Nondimensionalizing the force

Following the same logic as Section 4, divide the force by a natural scale. The dynamic pressure $q_\infty = \frac{1}{2}\rho V_\infty^2$ is the kinetic energy per unit volume of the oncoming flow and sets the scale of pressure forces the flow can plausibly produce; multiplying by a reference area S converts it to units of force. Defining

$$\boxed{C_L = \frac{L}{q_\infty S}, \quad C_D = \frac{D}{q_\infty S}, \quad C_M = \frac{M}{q_\infty S c}} \quad (5.1)$$

packages all the complicated shape- and condition-dependent behaviour into three dimensionless numbers. By the same Buckingham- π counting argument used to derive Re above, for a fixed shape and orientation these coefficients can only depend on the other dimensionless groups of the problem:

$$C_L = C_L(Re, Ma, \alpha), \quad C_D = C_D(Re, Ma, \alpha),$$

not on the physical size of the model – again the justification for scaled wind-tunnel testing.

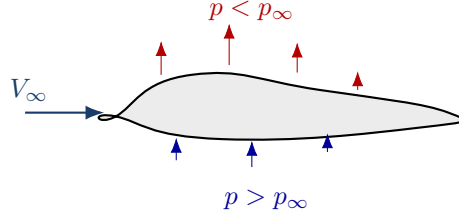


Figure 1: Pressure distribution around a lifting airfoil.

6 Momentum and Lift

6.1 An alternate route to the force: momentum flux

Instead of integrating pressure over the body's own surface, we can draw a large control volume around the airfoil and ask how much momentum flows in and out through its boundary – Newton's third law guarantees this must equal the reaction force on the body. Integrate Cauchy's equation (3.1) over the control volume V and apply the divergence theorem to convert the flux terms to a surface integral, exactly as in Section 2.2:

$$\boxed{\mathbf{F}_{\text{on fluid}} = \oint_{CS} \rho \mathbf{u} (\mathbf{u} \cdot \mathbf{n}) dA + \oint_{CS} p \mathbf{n} dA,} \quad (6.1)$$

and $\mathbf{F}_{\text{on body}} = -\mathbf{F}_{\text{on fluid}}$ by Newton's third law.

6.2 Simplifying far from the body

If we push the control surface far enough away, $p \rightarrow p_\infty$ (constant) everywhere on it, and a closed-surface integral of a constant times $\mathbf{n}$ is identically zero (this follows from the divergence theorem applied to a constant field), so the pressure term in (6.1) drops out at large distance. What's left is purely a momentum flux balance. Resolving the vertical component, with inflow at V_∞ and a small downward tilt (downwash) $w(y)$ at the exit plane,

$$L = - \int \rho u w dA \approx \rho V_\infty \int_{-\infty}^{\infty} w(y) dy. \quad (6.2)$$

The physical picture: the wing pushes air downward, so by Newton's third law the air pushes the wing upward; (6.2) makes that statement quantitative.

7 Circulation

7.1 Definition

Circulation is a loop integral of velocity around a closed curve C :

$$\boxed{\Gamma = \oint_C \mathbf{u} \cdot d\mathbf{l}.} \quad (7.1)$$

At each point on the loop we take the component of velocity along the direction of travel and add it up; if the flow tends to swirl in the same sense as the loop, contributions reinforce and Γ is large. Applying Stokes' theorem converts this loop integral into a surface integral of vorticity (defined in Section 8):

$$\Gamma = \int_A (\nabla \times \mathbf{u}) \cdot \mathbf{n} dA,$$

so circulation is the total spin passing through any surface bounded by the loop.

7.2 The Kutta–Joukowski theorem, derived

For ideal 2-D flow, lift and circulation are related by

$$\boxed{L' = \rho V_\infty \Gamma.} \quad (7.2)$$

To see where this comes from, represent the 2-D potential flow with a complex potential $w(\zeta) = \phi + i\psi$ (introduced properly in Section 9), whose derivative $dw/d\zeta = u - iv$ is the complex velocity. The Blasius formula states that the force on a body enclosed by a contour C can be computed by integrating the *square* of the complex velocity around that contour:

$$X - iY = \frac{i\rho}{2} \oint_C \left(\frac{dw}{d\zeta} \right)^2 d\zeta. \quad (7.3)$$

Far from the body the flow looks like uniform flow plus a point vortex of strength Γ (since any closed body with circulation looks like a vortex from far away), so

$$\frac{dw}{d\zeta} \rightarrow V_\infty - \frac{i\Gamma}{2\pi\zeta} + O(\zeta^{-2}).$$

Squaring this expression and using the residue theorem – only the $1/\zeta$ term in the squared expansion contributes to the contour integral, because all other powers of ζ integrate to zero around a closed loop – picks out a single surviving term. Carrying out that residue calculation in (7.3) gives $X = 0$ (zero drag, d'Alembert's paradox, a direct consequence of having neglected viscosity) and $Y = \rho V_\infty \Gamma$, which is exactly (7.2).

8 Vorticity

8.1 Definition

$$\boxed{\boldsymbol{\omega} = \nabla \times \mathbf{u},} \quad \boldsymbol{\omega} = 2\boldsymbol{\Omega}_{\text{local spin}}. \quad (8.1)$$

Vorticity is the curl of velocity, and it equals exactly twice the local angular velocity a tiny paddlewheel dropped into the flow at that point would spin at. Unlike circulation, which is defined relative to a chosen loop, vorticity is a purely local, pointwise quantity.

8.2 Deriving its transport equation

We can get an evolution equation for vorticity directly from momentum conservation: take the curl of both sides of (3.3). The curl of a gradient is always zero, so $\nabla \times (-\nabla p/\rho) = 0$ and $\nabla \times \mathbf{g} = 0$ (gravity is a constant vector) drop out immediately. For the convective term, the vector identity

$$\nabla \times (\mathbf{u} \cdot \nabla \mathbf{u}) = (\mathbf{u} \cdot \nabla) \boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \nabla) \mathbf{u}$$

(valid for incompressible flow) lets us rewrite that term in terms of vorticity instead of velocity. Taking the curl of the viscous term just commutes the curl and Laplacian operators, giving $\nu \nabla^2 \boldsymbol{\omega}$. Assembling the pieces:

$$\boxed{\frac{D\boldsymbol{\omega}}{Dt} = (\boldsymbol{\omega} \cdot \nabla) \mathbf{u} + \nu \nabla^2 \boldsymbol{\omega}.} \quad (8.2)$$

The first right-hand term is vortex stretching/tilting – existing vorticity being amplified or reoriented by velocity gradients (this term vanishes identically in 2-D, since there $\boldsymbol{\omega}$ points out of the plane and is perpendicular to every in-plane gradient of $\mathbf{u}$). The second term is ordinary viscous diffusion, the same operator as in Section 3.6. A direct consequence: if a flow starts irrotational ($\boldsymbol{\omega} = 0$) and is inviscid, (8.2) says $D\boldsymbol{\omega}/Dt = 0$, so it stays irrotational forever – meaning any vorticity that does appear in an otherwise-inviscid flow must originate at a viscous no-slip wall.

9 Potential Flow

9.1 From irrotational to a scalar potential

If $\boldsymbol{\omega} = \nabla \times \mathbf{u} = 0$ everywhere, a standard vector calculus identity guarantees $\mathbf{u}$ can be written as the gradient of some scalar ϕ , since the curl of any gradient is automatically zero:

$$\mathbf{u} = \nabla \phi.$$

This automatically satisfies irrotationality by construction, so we’ve traded a vector unknown $\mathbf{u}$ for a scalar unknown ϕ .

9.2 Applying continuity

We still need to satisfy mass conservation. Substituting $\mathbf{u} = \nabla \phi$ into the incompressible continuity equation (2.3):

$$\boxed{\nabla^2 \phi = 0.} \quad (9.1)$$

This is Laplace’s equation – linear, and the same equation that governs electrostatics and steady heat conduction, so a large existing body of solution techniques carries over directly.

9.3 The complex potential

In 2-D, incompressibility also lets us define a stream function ψ via $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$; substituting these into $\nabla \cdot \mathbf{u} = 0$ satisfies it identically for *any* smooth ψ , so mass conservation is automatic. Imposing irrotationality on this representation of $\mathbf{u}$ forces $\nabla^2\psi = 0$ as well. The pair (ϕ, ψ) satisfying the same Laplace equation and the Cauchy–Riemann relations is precisely what it means for

$$w(\zeta) = \phi(x, y) + i\psi(x, y), \quad \zeta = x + iy$$

to be an analytic function of the complex variable ζ . This is useful because analytic functions can be added together (linearity of Laplace’s equation) to build up complicated flows from simple building blocks: uniform flow $w = U\zeta$, a source $w = \frac{m}{2\pi} \ln \zeta$, a point vortex $w = \frac{-i\Gamma}{2\pi} \ln \zeta$, and a doublet $w = \frac{\kappa}{2\pi\zeta}$.

10 Bernoulli’s Equation

10.1 Deriving it from the momentum equation

Start from the steady, inviscid ($\mu = 0$) form of (3.3). Take its dot product with a small displacement $d\mathbf{l}$ along a streamline (i.e. integrate the momentum equation along the flow direction). The convective acceleration term simplifies using the identity

$$\mathbf{u} \cdot \nabla \mathbf{u} = \nabla \left(\frac{1}{2} |\mathbf{u}|^2 \right) - \mathbf{u} \times \boldsymbol{\omega},$$

and along a streamline $d\mathbf{l}$ is parallel to $\mathbf{u}$, so the $\mathbf{u} \times \boldsymbol{\omega}$ term, being perpendicular to $\mathbf{u}$, contributes nothing when dotted with $d\mathbf{l}$. Writing gravity as $\mathbf{g} = -\nabla(gz)$ (a conservative force), the whole dotted equation collapses to the statement that the gradient of a single combined quantity is zero along the streamline:

$$\nabla \left(\frac{1}{2} \rho V^2 + p + \rho gz \right) \cdot d\mathbf{l} = 0 \quad \implies \quad \boxed{p + \frac{1}{2} \rho V^2 + \rho gz = \text{const along a streamline.}} \quad (10.1)$$

If the flow is also irrotational ($\boldsymbol{\omega} = 0$), the dropped term $\mathbf{u} \times \boldsymbol{\omega}$ is zero *everywhere*, not just along one streamline, so the identity $\mathbf{u} \cdot \nabla \mathbf{u} = \nabla(\frac{1}{2} V^2)$ holds globally and the constant in (10.1) is the same on every streamline in the flow, not just constant along each one individually.

10.2 What it is and isn’t

Equation (10.1) is a first integral of the momentum equation under specific assumptions (steady, inviscid, incompressible) – it is a *consequence* of Newton’s second law under those restrictions, not an independent physical law, and shouldn’t be applied where those assumptions fail (strongly viscous, separated, or compressible flow). It’s also worth noting that using the true velocity field from a correctly solved potential-flow problem, Bernoulli gives the right pressure difference across a wing – but the popular “equal transit time” explanation for why the upper-surface flow is faster is not itself correct; the actual velocity field must come from solving (9.1) with the Kutta condition, not assumed a priori.

11 The Joukowski Transformation

11.1 The mapping

$$\boxed{z = \zeta + \frac{a^2}{\zeta}}. \quad (11.1)$$

The idea: potential flow around a circular cylinder has been solvable in closed form since the 19th century, and because this map is conformal (angle-preserving), any solution of Laplace's equation (9.1) in the simple ζ -plane maps to another valid solution in the complicated-looking z -plane airfoil geometry. So instead of solving a hard boundary-value problem directly on an airfoil shape, we solve the easy circular-cylinder problem and let the map carry the solution over.

11.2 Building the flow around an offset circle

Place a circle of radius $R \geq a$ centered at $\zeta_0 = -\epsilon + i\beta_0$ (a small offset from the origin produces camber after mapping). Superposing uniform flow at angle of attack α , a doublet (giving the circle its shape), and a point vortex of strength Γ centered at ζ_0 – exactly the building-block potentials introduced in Section 9.3 – gives the complex potential

$$w(\zeta) = V_\infty \left[(\zeta - \zeta_0)e^{-i\alpha} + \frac{R^2}{\zeta - \zeta_0}e^{i\alpha} \right] - \frac{i\Gamma}{2\pi} \ln(\zeta - \zeta_0). \quad (11.2)$$

11.3 Applying the Kutta condition

Physically, the flow must leave the sharp trailing edge smoothly rather than wrapping around it – the Kutta condition. The trailing edge is the point $\zeta = a$ on the circle where the mapping (11.1) itself becomes singular, since $dz/d\zeta = 1 - a^2/\zeta^2 = 0$ there; a finite, well-behaved velocity in the physical z -plane at that point therefore *requires* the complex velocity $dw/d\zeta$ in the ζ -plane to vanish there too, otherwise the mapped velocity would blow up. Imposing $dw/d\zeta = 0$ at $\zeta = a$ in (11.2) and solving for Γ fixes the circulation uniquely – it is no longer a free parameter but is determined by the requirement of smooth trailing-edge flow:

$$\boxed{\Gamma = -4\pi V_\infty R \sin(\alpha + \beta_0)}. \quad (11.3)$$

11.4 Recovering thin-airfoil theory

Substitute (11.3) into the Kutta–Joukowski result (7.2). For a thin, nearly-flat airfoil, $R \approx a$ and $\beta_0 \approx 0$, and expanding $\sin \alpha \approx \alpha$ for small angle of attack gives

$$\boxed{c_l = 2\pi(\alpha - \alpha_{L0}) \approx 2\pi\alpha}, \quad \frac{dc_l}{d\alpha} = 2\pi \text{ rad}^{-1}. \quad (11.4)$$

This 2π lift-curve slope, obtained here from the exact Joukowski-circle solution, is the same result independently obtained from thin-airfoil vortex-sheet theory, and it matches low- α experimental data well until viscous separation (stall) intervenes.

12 Viscosity and No-Slip

12.1 The boundary condition and its consequence

Molecules of fluid immediately touching a solid surface adhere to it, so relative to the wall,

$$\boxed{\mathbf{u}|_{\text{wall}} = 0.} \quad (12.1)$$

Consider what this forces to happen nearby: the fluid velocity must transition from 0 at the wall up to the free-stream value U over some finite distance Δy . From (3.5), the shear stress this produces scales as $\tau \sim \mu \Delta u / \Delta y$; if Δy turns out to be small, that stress can be substantial even for a small-viscosity fluid like air. Working out how small Δy actually is, and why, is the subject of the boundary layer, next.

13 Boundary Layers

13.1 Scaling estimate

Prandtl's key insight was that at high Reynolds number the viscous term in (3.3) can only matter in a thin layer near the wall, because that's the only place velocity gradients are steep enough to make it compete with inertia. We can estimate the thickness δ of that layer by balancing the two: the convective term scales as $U \cdot U/x$ (velocity times its own streamwise gradient), and the viscous term scales as $\nu \cdot U/\delta^2$ (viscosity times the curvature of the velocity profile across the thin layer). Setting these equal, since both are the same order inside the boundary layer by definition:

$$U \frac{U}{x} \sim \nu \frac{U}{\delta^2} \quad \Rightarrow \quad \boxed{\delta \sim \sqrt{\frac{\nu x}{U}} \quad \Leftrightarrow \quad \frac{\delta}{x} \sim Re_x^{-1/2}.} \quad (13.1)$$

The square-root growth is the signature of a diffusive process: the wall "drags" its neighbours along via viscosity, and diffusive spreading distances always grow with the square root of elapsed time – here, downstream distance x/U stands in for time.

13.2 The Blasius similarity solution

We can do better than a scaling estimate and solve the boundary-layer equations exactly for a flat plate with zero pressure gradient. The trick is to guess that the velocity profile has the same *shape* at every streamwise station once rescaled by the local boundary-layer thickness – a similarity solution. Define the similarity variable $\eta = y\sqrt{U/(\nu x)}$ (this rescales y by exactly the $\delta(x)$ we found in (13.1)) and a stream function $\psi = \sqrt{\nu U x} f(\eta)$, so that $u/U = \partial\psi/\partial y \cdot (1/U) = f'(\eta)$. Substituting this ansatz into the boundary-layer form of (3.3) converts the original PDE in (x, y) into a single ODE in η alone – the whole point of the similarity substitution:

$$\boxed{2f''' + ff'' = 0,} \quad f(0) = f'(0) = 0, \quad f'(\infty) = 1. \quad (13.2)$$

The boundary conditions are no-slip ($f'(0) = 0$), no flow through the wall ($f(0) = 0$), and matching the free stream far away ($f'(\infty) = 1$). This ODE has no closed-form solution but is solved numerically to high precision, giving $f''(0) \approx 0.332$; since $f''(0)$ is proportional to the wall velocity gradient, this directly gives the wall shear stress:

$$\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} = 0.332 \rho U^2 Re_x^{-1/2}. \quad (13.3)$$

13.3 Integral thicknesses and the Kármán equation

Solving the full boundary-layer PDE (or the Blasius ODE) is often more detail than needed. A cheaper approach integrates the momentum equation across the boundary layer once, trading a PDE for an ODE in x alone, at the cost of needing an assumed velocity-profile shape. First define two integral thickness measures that summarize the profile:

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) dy, \quad \theta = \int_0^\infty \frac{u}{U} \left(1 - \frac{u}{U}\right) dy, \quad (13.4)$$

δ^* measures how far the outer streamlines are displaced outward by the slowed-down near-wall fluid, and θ measures the deficit in momentum flux caused by the boundary layer, relative to inviscid flow at speed U . Integrating the boundary-layer momentum equation over y from the wall to the free stream, and using integration by parts together with the definitions in (13.4) to collapse the result into these two thicknesses, gives the Kármán momentum-integral equation:

$$\boxed{\frac{d\theta}{dx} + \frac{\theta}{U}(2 + H)\frac{dU}{dx} = \frac{\tau_w}{\rho U^2}}, \quad H = \frac{\delta^*}{\theta}. \quad (13.5)$$

This single ODE for $\theta(x)$ is the basis of integral (closure) methods – such as Head’s method – used whenever solving the full boundary-layer PDE directly is too expensive, by supplying an additional empirical relation for H and τ_w in terms of θ .

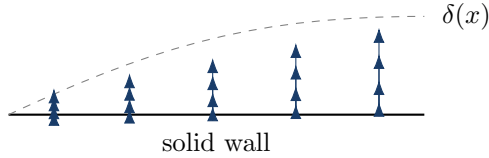


Figure 2: Growing laminar boundary layer, $u(y)$ from 0 at the wall to U at $\delta(x)$.

14 Pressure Gradients and Separation

14.1 Definition

$$\boxed{\tau_w = \mu \left. \frac{\partial u}{\partial y} \right|_w}, \quad \text{separation: } \tau_w = 0. \quad (14.1)$$

Separation happens when near-wall fluid, already slowed by friction, runs out of momentum to keep advancing against a rising pressure and locally reverses direction; the point where that reversal begins is exactly where the wall shear passes through zero.

14.2 Why an adverse pressure gradient causes an inflection point

We can connect this directly to the momentum equation. Evaluate the full (not boundary-layer-simplified) momentum equation at the wall itself, $y = 0$, where the no-slip condition forces $u = v = 0$. Every term involving u or v then drops out except the pressure and the wall-normal viscous term, leaving a compatibility condition:

$$\mu \left. \frac{\partial^2 u}{\partial y^2} \right|_w = \frac{dp}{dx}. \quad (14.2)$$

If $dp/dx > 0$ (adverse gradient), (14.2) says the velocity profile has positive curvature right at the wall. But far from the wall the profile must curve the other way to blend smoothly into the free stream at $u \rightarrow U$. A profile that curves one way near the wall and the other way farther out must have an inflection point somewhere in between – and by Rayleigh’s inflection-point criterion, an inflectional velocity profile is inviscidly unstable. This links separation directly to the transition-to-turbulence discussion in Section 16.

15 Flow Instability and Vortex Shedding

$$\boxed{St = \frac{fD}{U}}. \quad (15.1)$$

Two shear layers separating off the top and bottom of a bluff body are each individually unstable; any small asymmetry gets amplified, one side sheds a vortex first, that changes the local pressure field enough to trigger shedding from the other side, and the process locks into a regular alternating rhythm at frequency f . For a circular cylinder, $St \approx 0.2$ stays nearly constant over a wide range, $300 \lesssim Re \lesssim 2 \times 10^5$.



Figure 3: Von Kármán vortex street; oscillating lift at $f = StU/D$.

16 Reynolds Decomposition and the Closure Problem

16.1 Splitting velocity into mean and fluctuation

Once a flow is turbulent, tracking every chaotic instantaneous detail is neither tractable nor useful. Instead, decompose the velocity at every point into a time-averaged mean and a fluctuation about that mean,

$$u_i = U_i + u'_i, \quad \overline{u'_i} = 0$$

by construction (the fluctuation is defined as whatever is left over after subtracting the mean).

16.2 Substituting into the momentum equation

Substitute this decomposition into (3.3) and take the time average of the whole equation. Time-averaging commutes with $\partial/\partial t$ and ∇ , so the linear terms (pressure, viscous, gravity) are unchanged in form, just

with u replaced by U . The convective term is where something new happens, because it's quadratic in velocity and the average of a product is not, in general, the product of the averages:

$$\overline{u_j \frac{\partial u_i}{\partial x_j}} = \overline{(U_j + u'_j) \frac{\partial (U_i + u'_i)}{\partial x_j}} = U_j \frac{\partial U_i}{\partial x_j} + \frac{\partial \overline{u'_i u'_j}}{\partial x_j},$$

where the cross terms linear in a single fluctuation vanish upon averaging ($\overline{u'_i} = 0$), but the $\overline{u'_i u'_j}$ term survives because a fluctuation correlated with itself doesn't average to zero. Substituting this back in gives the Reynolds-averaged Navier–Stokes (RANS) equation:

$$\boxed{\rho \left(\frac{\partial U_i}{\partial t} + U_j \frac{\partial U_i}{\partial x_j} \right) = -\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j} \left(\mu \frac{\partial U_i}{\partial x_j} - \rho \overline{u'_i u'_j} \right).} \quad (16.1)$$

16.3 Why this isn't a closed system

Equation (16.1) looks just like the original Navier–Stokes equation for the mean flow U_i , except for the extra term $-\rho \overline{u'_i u'_j}$, the Reynolds stress. But this term is a genuinely new unknown – it's not expressible in terms of U_i and p alone – and being symmetric it has 6 independent components in 3-D. We now have more unknowns than equations: this is the closure problem, and it's why turbulence modelling is necessary at all.

17 The Turbulent Energy Cascade

17.1 Where the small scales come from

Kinetic energy is injected at large scales set by the geometry (wake width, wing chord). Those large eddies are themselves unstable and break into smaller ones, transferring energy down the scale range with little loss at each step, until eddies become small enough that viscosity finally becomes efficient at converting their motion to heat. The scales at which that dissipation happens, the Kolmogorov scales, are estimated purely from dimensional analysis using the two quantities available at the smallest scales – the dissipation rate ε and kinematic viscosity ν – combined into the only combination with units of length, time, and velocity respectively:

$$\boxed{\eta = \left(\frac{\nu^3}{\varepsilon} \right)^{1/4}, \quad \tau_\eta = \left(\frac{\nu}{\varepsilon} \right)^{1/2}, \quad u_\eta = (\nu \varepsilon)^{1/4}.} \quad (17.1)$$

17.2 Estimating the range of active scales

At the large-eddy end, dimensional reasoning gives $\varepsilon \sim u_0^3/\ell$ (energy per unit mass per unit time, built from the large-eddy velocity u_0 and size ℓ). Substituting this into (17.1) and forming the ratio of large to small length scales:

$$\boxed{\frac{\ell}{\eta} \sim Re^{3/4}, \quad \frac{\tau_0}{\tau_\eta} \sim Re^{1/2}.} \quad (17.2)$$

A 3-D simulation needs to resolve this whole range of scales in each of three directions, so the total number of grid points required scales as $(\ell/\eta)^3 \sim Re^{9/4}$ – this is precisely the cost argument that makes fully

resolved DNS prohibitively expensive at high Reynolds number.

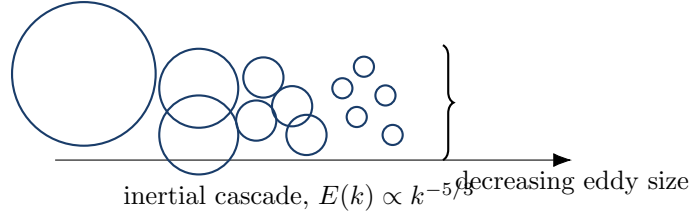


Figure 4: Kolmogorov energy cascade.

18 Why CFD Is Necessary

Equations (3.3)/(16.1) admit closed-form solutions only for the small set of idealized geometries handled analytically in Sections 9–11. For arbitrary geometry there is no shortcut: we must replace the exact continuum PDE with a finite-dimensional numerical approximation that a computer can actually solve, at the cost of exactness – the subject of the rest of this handout.

19 Finite-Volume Method

19.1 Deriving the discrete conservation statement

Start from a generic conservation law in flux form, $\partial U / \partial t + \nabla \cdot \mathbf{F} = S$, and integrate it over a single mesh cell V_c , then apply the divergence theorem to the flux term exactly as we did for mass conservation back in Section 2.2:

$$\boxed{\frac{d}{dt} \int_{V_c} U dV + \sum_{\text{faces } f} \mathbf{F}_f \cdot \mathbf{n}_f A_f = \int_{V_c} S dV,} \quad (19.1)$$

where the continuous surface integral has been replaced by a finite sum over the cell’s discrete faces – the only approximation introduced so far is representing the surface integral as a sum over flat faces.

19.2 Why this guarantees exact conservation

Here’s the key structural property: the flux $\mathbf{F}_f \cdot \mathbf{n}_f$ computed by one cell on a face it shares with a neighbour is used, with the opposite sign, as that neighbour’s flux across the same face (what leaves one cell must be exactly what enters the other). Summing (19.1) over every cell in the mesh, all of these shared internal face terms cancel in pairs, leaving only the fluxes on the domain’s outer boundary:

$$\sum_c \left(\frac{d}{dt} \int_{V_c} U dV \right) = \int_{\partial\Omega} (-\mathbf{F} \cdot \mathbf{n}) dA + \int_{\Omega} S dV.$$

This is exact global conservation, regardless of mesh coarseness or irregularity, and is the central reason the finite-volume method is preferred for CFD. Discretizing (19.1) for every cell and unknown turns the PDE into a large linear algebra problem:

$$\boxed{\mathbf{A}\mathbf{x} = \mathbf{b}.}$$

20 Numerical Discretization

20.1 Deriving the truncation error of a finite-difference formula

A computer can only evaluate a function at discrete points, so any numerical derivative needs a finite spacing Δx . To find out how accurate a given formula is, Taylor-expand the function about point i :

$$\begin{aligned} u_{i+1} &= u_i + \Delta x u'_i + \frac{1}{2} \Delta x^2 u''_i + \frac{1}{6} \Delta x^3 u'''_i + O(\Delta x^4), \\ u_{i-1} &= u_i - \Delta x u'_i + \frac{1}{2} \Delta x^2 u''_i - \frac{1}{6} \Delta x^3 u'''_i + O(\Delta x^4). \end{aligned}$$

Subtracting the second from the first, the even-order terms (u_i, u''_i) cancel identically, leaving

$$u_{i+1} - u_{i-1} = 2\Delta x u'_i + \frac{1}{3} \Delta x^3 u'''_i + O(\Delta x^5),$$

and dividing by $2\Delta x$:

$$\frac{u_{i+1} - u_{i-1}}{2\Delta x} = u'_i + \frac{\Delta x^2}{6} u'''_i + O(\Delta x^4). \quad (20.1)$$

The leading error term is $O(\Delta x^2)$, so this central-difference formula is second-order accurate: halving Δx cuts the error by roughly a factor of four.

20.2 The CFL condition, from a stability argument

For explicit time-marching of a convective term with wave speed c , information physically propagates a distance $c\Delta t$ in one time step. The numerical scheme, meanwhile, can only "see" information from neighbouring grid points a distance Δx away. If the physical domain of dependence ($c\Delta t$) outruns the numerical one (Δx), the scheme is trying to use information it doesn't actually have access to, and the discrete solution diverges from the true one – instability. Requiring the numerical domain of dependence to contain the physical one gives the stability requirement:

$$\boxed{\text{CFL} = \frac{c\Delta t}{\Delta x} \leq C_{\max}}, \quad (20.2)$$

with $C_{\max} = 1$ for simple explicit schemes. This is why Δt must shrink whenever the mesh is refined in an explicit solver, and why implicit schemes (which use future, not just current, neighbour values, and are unconditionally stable) are often preferred on fine meshes despite requiring a linear solve every step.

21 Convergence Is Not Accuracy

Total error is the sum of several distinct, independent contributions: spatial discretization $O(\Delta x^p)$ (Section 20.1), temporal discretization $O(\Delta t^q)$, iterative/residual error $\|\mathbf{r}_k\| = \|\mathbf{b} - \mathbf{A}\mathbf{x}_k\| \rightarrow 0$ as the linear solver iterates, and modelling error (turbulence closure, boundary conditions). Driving the residual to machine-zero – what "converged" usually means in practice – controls only the third of these; the mesh could still be too coarse, or the turbulence model wrong for the flow, and the solver would happily converge to the wrong answer.

A converged numerical solution is not necessarily a correct physical solution.

22 DNS, LES, and RANS

22.1 DNS

Resolve all scales down to η directly, with no turbulence model. From (17.2), the grid-point cost scales as $Re^{9/4}$ and the number of time steps needed scales as $Re^{1/2}$, so total cost scales roughly as $Re^{11/4}$ – explosively expensive at engineering Reynolds numbers.

22.2 LES

Apply a spatial filter of width Δ to the velocity field, $\overline{u}_i(\mathbf{x}) = \int G_\Delta(\mathbf{x} - \mathbf{x}') u_i(\mathbf{x}') d\mathbf{x}'$, which resolves eddies larger than Δ directly and removes (filters out) smaller ones. Filtering (3.3) the same way we time-averaged it in Section 16 produces an equation structurally identical to (16.1), but with a subgrid-scale stress $\tau_{ij}^{sgs} = \overline{u_i u_j} - \overline{u_i} \overline{u_j}$ in place of the Reynolds stress, representing only the effect of the small, filtered-out scales rather than all turbulent fluctuations. Common closures again use an eddy-viscosity analogy, e.g. Smagorinsky: $\tau_{ij}^{sgs} \approx -2\nu_t \overline{S}_{ij}$, $\nu_t = (C_s \Delta)^2 |\overline{S}|$.

22.3 RANS

Solve only the fully time-averaged equation (16.1), modelling the *entire* Reynolds stress rather than just a subgrid piece of it. Cheapest of the three approaches, and the workhorse of everyday engineering CFD.

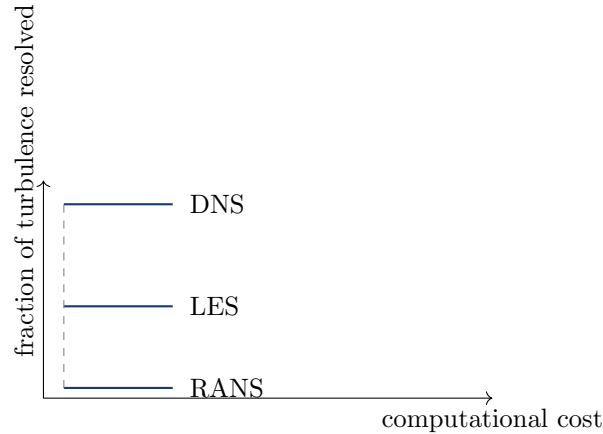


Figure 5: Cost vs. fidelity trade-off.

23 Eddy-Viscosity Models

23.1 The Boussinesq closure hypothesis

Ordinary molecular viscosity relates viscous stress to mean velocity gradient via $\tau = \mu \partial u / \partial y$ (Section 3.6). The Boussinesq hypothesis proposes modelling the unknown Reynolds stress the same way – as if turbulent

mixing behaved like an enhanced viscosity – introducing a new, flow-dependent turbulent viscosity μ_t in place of the fixed molecular μ :

$$\boxed{-\rho \overline{u'_i u'_j} \approx 2\mu_t S_{ij} - \frac{2}{3}\rho k \delta_{ij}}, \quad S_{ij} = \frac{1}{2} \left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i} \right), \quad k = \frac{1}{2} \overline{u'_i u'_i}. \quad (22.1)$$

The $-\frac{2}{3}\rho k \delta_{ij}$ term is needed so that contracting both sides ($i = j$) correctly reproduces $2k$ on the left, since $S_{ii} = 0$ for incompressible flow. This is a deliberate modelling analogy, not something derivable from first principles the way (3.3) was – μ_t isn't a material property, it varies with position and flow condition, and must itself be computed.

23.2 Deriving the exact turbulent kinetic energy equation

Before modelling μ_t , we can derive an *exact* transport equation for k directly from the Navier–Stokes equations, the same way we derived (16.1): take the exact (unmodelled) equation governing each individual Reynolds-stress component $\overline{u'_i u'_j}$, set $i = j$, and halve the result (since $k = \frac{1}{2} \overline{u'_i u'_i}$). This bookkeeping exercise produces

$$\boxed{\frac{Dk}{Dt} = \underbrace{-\overline{u'_i u'_j} \frac{\partial U_i}{\partial x_j}}_{\text{production } P_k} - \underbrace{\varepsilon}_{\text{dissipation}} + \nabla \cdot [(\nu + \nu_t/\sigma_k) \nabla k]}. \quad (22.2)$$

Substituting the Boussinesq closure (22.1) into the production term P_k gives $P_k = 2\nu_t S_{ij} S_{ij} = \nu_t |S|^2$: production of turbulent kinetic energy scales with the square of the mean strain rate, the same S_{ij} that (via Bradshaw's Richardson-number correction) also governs curvature-driven production over curved walls.

24 Common RANS Models

To use (22.1)–(22.2) we still need an equation for μ_t itself. Dimensionally, μ_t has units of (density) $\times$ (velocity) $\times$ (length), and k already supplies a velocity scale ($\sqrt{k}$); pairing it with a dissipation-based length or time scale closes the dimensions. The standard k – ε model adds a second, modelled transport equation for ε itself (built by analogy with the exact k -equation (22.2), since deriving an exact ε -equation produces terms that themselves need modelling):

$$\frac{D\varepsilon}{Dt} = C_{\varepsilon 1} \frac{\varepsilon}{k} P_k - C_{\varepsilon 2} \frac{\varepsilon^2}{k} + \nabla \cdot [(\nu + \nu_t/\sigma_\varepsilon) \nabla \varepsilon], \quad \mu_t = \rho C_\mu \frac{k^2}{\varepsilon}, \quad (23.1)$$

with the eddy viscosity now fixed dimensionally by $[k^2/\varepsilon] = (\text{length}^2/\text{time}^2)^2/(\text{length}^2/\text{time}^3) = \text{length}^2/\text{time}$, matching $[\mu_t/\rho]$, and calibrated constants $C_\mu \approx 0.09$, $C_{\varepsilon 1} \approx 1.44$, $C_{\varepsilon 2} \approx 1.92$. This model performs well away from walls but ε behaves poorly as the wall is approached, requiring special near-wall treatment. The k – ω family instead transports the specific dissipation rate $\omega = \varepsilon/(C_\mu k)$,

$$\mu_t = \rho \frac{k}{\omega}, \quad (23.2)$$

which integrates more robustly all the way to the wall but is more sensitive to the free-stream value specified far away. SST k – ω blends (23.1) and (23.2) using a function $F_1(y^+)$ that equals 1 near walls and 0 in the free stream, combining the near-wall accuracy of k – ω with the free-stream robustness of k – ε .

None of the model equations in this section is derived from first principles the way (3.3) or (16.1) were; each is calibrated against experimental and DNS data for benchmark flows, and accuracy degrades when applied well outside those calibration conditions.

25 Verification and Validation

Verification asks a purely numerical question: does the discrete solution $\mathbf{x}_h$ converge to the exact solution of the *chosen* PDE and closure as $\Delta x, \Delta t \rightarrow 0$ and $\|\mathbf{r}_k\| \rightarrow 0$? Validation asks a physical question: does that chosen PDE and closure actually represent reality, checked only by comparison against experiment? Neither answer implies the other.

A converged, verified simulation can still fail validation if the turbulence model or boundary conditions do not match the real physics.

26 CFD as a Design Tool

26.1 Setting up the variational problem

Design asks the inverse of prediction: given a desired flow behaviour, what geometry produces it? Formalize this by defining an objective functional over candidate shapes $y(x)$,

$$J[y] = \int_{x_0}^{x_1} F(x, y, y') dx,$$

where F might represent local drag, pressure loss, or (as in curvature-induced turbulence production) a function of wall curvature. We want the particular $y(x)$ that minimizes J .

26.2 Deriving the Euler–Lagrange equation

Consider perturbing the candidate solution by a small admissible variation, $y \rightarrow y + \epsilon \eta(x)$, where $\eta(x_0) = \eta(x_1) = 0$ keeps the endpoints fixed. If y truly minimizes J , then J can't decrease to first order under *any* such perturbation, so the derivative of J with respect to ϵ must vanish at $\epsilon = 0$:

$$\left. \frac{d}{d\epsilon} J[y + \epsilon \eta] \right|_{\epsilon=0} = 0.$$

Differentiating under the integral sign (chain rule on F):

$$0 = \int_{x_0}^{x_1} \left(\frac{\partial F}{\partial y} \eta + \frac{\partial F}{\partial y'} \eta' \right) dx.$$

The second term still has η' in it, which we'd like to eliminate so that both terms are multiplied by plain η . Integrate that term by parts:

$$\int_{x_0}^{x_1} \frac{\partial F}{\partial y'} \eta' dx = \left[\frac{\partial F}{\partial y'} \eta \right]_{x_0}^{x_1} - \int_{x_0}^{x_1} \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \eta dx,$$

and the boundary term vanishes because $\eta(x_0) = \eta(x_1) = 0$ by construction. Substituting back:

$$0 = \int_{x_0}^{x_1} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] \eta \, dx.$$

This integral must vanish for *every* admissible $\eta(x)$. The fundamental lemma of the calculus of variations says the only way that can happen is if the bracketed term is itself identically zero (if it were nonzero anywhere, we could choose an η concentrated there to make the integral nonzero – a contradiction). This localization argument is the same logic used in Section 2.3 to go from an integral statement to a pointwise one. The result:

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.} \quad (25.1)$$

26.3 Extending to curvature-dependent objectives

If F depends on curvature as well, $F(x, y, y', y'')$, repeat exactly the same variational argument but integrate the η'' term by parts *twice* instead of once to eliminate both derivatives of η . The boundary terms vanish under the additional assumption $\eta'(x_0) = \eta'(x_1) = 0$ (fixed slope at the endpoints), and the same localization argument gives the Euler–Poisson equation:

$$\boxed{\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) + \frac{d^2}{dx^2} \left(\frac{\partial F}{\partial y''} \right) = 0.} \quad (25.2)$$

This is the relevant equation whenever the design objective depends explicitly on wall curvature $\kappa \approx y''$, as in curvature-induced turbulence production. Adding equality constraints (e.g. a fixed boundary-layer state at the outlet) via Lagrange multiplier functions – costates – converts (25.1)/(25.2) into a larger, coupled two-point boundary value problem, which is exactly the structure of Pontryagin’s minimum principle: a state equation and a costate equation replacing the single Euler–Lagrange ODE.

27 A Useful Measure of Unsteadiness

27.1 Why not just average the fluctuation

Given a fluctuating signal $u(t) = U + u'(t)$, averaging $u'(t)$ directly gives zero by definition (that’s how U was chosen), so it can’t be used as a measure of *how much* the signal fluctuates. Squaring first removes the sign so positive and negative excursions no longer cancel, then averaging and taking a square root returns to the original units:

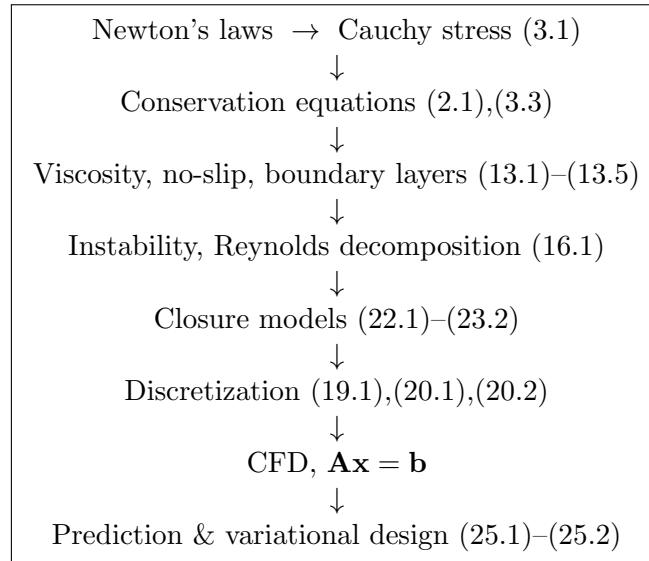
$$\boxed{u_{\text{rms}} = \sqrt{\langle (u - U)^2 \rangle} = \sqrt{\overline{u'^2}}.} \quad (26.1)$$

For isotropic turbulence, where the three fluctuation components are statistically identical, $\overline{u'^2} = \overline{v'^2} = \overline{w'^2}$, so the definition of k from (22.1) reduces to

$$k = \frac{1}{2} \left(\overline{u'^2} + \overline{v'^2} + \overline{w'^2} \right) = \frac{3}{2} u_{\text{rms}}^2,$$

showing u_{rms} and k carry the same information in that limiting case.

28 The Central Picture



**The fundamental question is not “Can we simulate it?”
It is “Do we understand what the simulation is actually telling us?”**